Arab Academy for Science and Technology and Maritime Transport

College of Computing and Information Technology

ECE4302 – Advanced Networks

PROJECT

Internet of Things (IoT) is a new predominant technology for this advanced world. This technology can change the lifestyle people lead. IoT can be described as a network of physical objects connected through the internet. Physical objects could be anything that contains embedded electronics, software, sensors, etc. with the internet. Using the IP addresses, those smart objects can exchange data among the network and can make a decision.

Students are requested to implement an IOT-based project with the help of embedded electronics and sensors:

PROJECT PHASES

Project consists mainly of 3 phases:

1. Arduino code to read the sensor's data
2. Server sends the data to a Webpage or mobile app to display the result
3. The connection between step 1 and step 2

LIST OF PROJECTS

1. **Weather Forecast System**

   It should allow people to directly check the weather stats online without the need of a weather forecasting agency. System uses a collection of sensors to monitor weather and provide live reporting of the weather statistics. Also, the system should allow users to set alerts for particular instances, the system provides alerts to user if the weather parameters cross those values.

2. **Health Monitoring System**

   This System should be able to monitor a patient's basic health signs as well as the room condition. In this system, five sensors can be used to capture the data from hospital environment: heart beat sensor, body temperature sensor, room temperature sensor, CO sensor, and CO2 sensor. The condition of the patients is conveyed via a portal to medical staff, where they can process and analyze the current situation of the patients.

3. **Street Light Monitoring System**

   This system consists of smart lights that automatically turns on at desired intensity based on amount of lighting needed. Also, it should detect if any of the lights has a fault, it should then automatically flag that a light is faulty and this data is sent over to the IOT monitoring system so that action can be taken to fix it.

4. **Smart home lighting**

   This system should be able to monitor the lightning of a house in real-time. Lights should be turned on when there's movement in its range. Users should be able to monitor the lights, turn them on/ off and get notified when a light turns on due to a movement on the house.

5. **Smart garage door**

   This should system should be able to close/ open a garage door remotely. It should also view its state (open/ closed) and provide an alarming system that has the ability to notify the owner if someone tries to access the garage door.

6. **Liquid level monitoring system**

   This system should be able to inform the users about the level of the liquid in a container and prevent it from overflowing. A graphical view of the containers and highlights the liquid level in color in order to show the level of liquid should be implemented. The system should also notify the users when the level of liquid collected crosses the set limit.

7. **Smart dustbin**

   This system should be able to detect human gesture and open automatically without anyone needing to press its lever. Also, the dustbin should be able to measure the level of garbage in the bin and automatically detects if it is about to fill up. The system should also be able to monitor a number of dustbins and highlight the filled ones and notify the user.

8. **IOT Based Air Pollution Monitor System**

   This system should be able to monitor the air quality by monitoring the amount of harmful gases like CO_2, smoke, alcohol, benzene and NH_3.Also, It should trigger an alarm when the air quality goes down beyond a certain level.

9. **Smart parking system**

   This system should be able to sense parking slot occupancy, and users could check online if there's a space available. Also, it should be able to sense if a vehicle has arrived on gate for automated gate opening.

10. **Fire Detection System**

    The system should be able to notify the user and activate fire sprinkles if a fire breaks out. Also, it should be able to notify authorities if the user doesn't respond within a specific period time.

11. **Gas Leak Detector**

    This system should be able to detect any gas leakage early, it should continuously monitor the level of LPG gas present in the air. While monitoring, if the value of LPG gas in air exceeds a certain limit, user should be notified and the gas valve should be closed immediately.

12. **Smart Agricultural System**

    This system should allow user to automate irrigation and control it remotely. The system should be able to open the irrigation valve automatically on times set by the user. Also, the user should be able to take control of the valves at any time.

13. **Refrigerator Tray**

    This system should be able to count a certain product(s) on a tray in a fridge. Also, it should notify the user to buy more if the count decreases to a certain threshold.

14. **Aircraft State Monitoring System**

    This system can measure the altitude, orientation, and atmospheric pressure of an aircraft/drone while in-flight. It allows Ground control to view current state of these values at all times and get warning if any value goes beyond normal conditions.

15. **Smart Pet Feeding System**

    This system allows trained dogs/animals to dispense food from a container by touch/button press. The system lets users keep track of how many times the dispenser was activated and how much food is left in container. Users should be notified if the container needs to be refilled with food.

16. **Smart Baby Monitoring System**

    This system monitors babies while sleeping in their cradles. The system can detect when the baby cries and alert the parents, allowing the parents to activate a rocking motion to help the baby sleep again. Parents can remotely control the speed of rocking motor and stop it at any time.

17. **Remote Security System**

    This system can be installed at doors in a high-security area. The security system should feature a PIR sensor that detects motion, a buzzer that goes off when motion is detected, and a motor that allows locking/unlocking of door. Once motion is detected, the guards are notified. They can turn off buzzer, and lock/unlock the door remotely.

18. **Weighing scale**

    Using this project, we can measure and monitor weights from any part of the world. The load cell (strain gauge) used here is of 5 kg. A to D converter converts the physical weight to corresponding digital signals and sends it to the owner with the aid of Arduino.

19. **Food spoilage detector:**

    This project regularly tracks and monitors the quality of food in our household. The gas sensors sense the smell of spoiled food and relay a message to the Arduino. Arduino flashes the message on the LCD display that the food is spoiled. Along with this, the owner also receives a message for the same.

20. **Water quality monitoring:**

    This project regularly monitors and tracks the pH value of the water and fitness of water. The range of pH between 7.2 and 7.8 is considered fit for drinking purposes. The pH value of water is periodically updated to the owner over the internet.

21. **Password Security Lock System:**

    We are going to make a Password Security Lock System using Arduino, Keypad & Servo. Security is becoming a major concern these days as theft is on the rise. So digital password lock can easily secure your home or locker. It only opens your door when the correct password is entered.

22. **Smart Home Entrance System**

    This system should be able to check for authorized guests using a camera that gets triggered upon pressing a button then checks:
    - If the guest is known to the system (using face recognition), his name is sent to the homeowner.
    - Else, the guest's picture is sent instead.
    At all cases, the homeowner decides whether he/she wants to open the door for the guest or not.

23. **Smart Room Cooling System**

    This system should be able to track the room's temperature and cool it using fans mounted to the room's ceiling. The system has two modes left to the user's desire: Automatic and Manual. Automatic: Feedback about the room status is sent to the user along with recommendations about the fan level (Off, 1, 2, 3, etc.) suitable for the current room temperature. Manual: The user runs the fan on the level he wishes disregarding the room temperature.

24. **Surveillance Automobile**

    This system includes a car (e.g. RC Car) controlled wirelessly (e.g. using Bluetooth). A camera is to be mounted on the car and feedback should be sent from the camera and tracked by a mobile/web application.

25. **Smart Alarm System**

    This system should be able to check for users' authorization of entering a hall or room. The system includes a keypad/mobile input and an LCD display. The user enters the password, if the user passes password authorization, the door is unlocked, and he/she is allowed to enter. Else, if the user doesn't pass authorization and tries to forcefully enter and opens the door, a buzzer sounds, alarm lights are fired and a notification is sent to the host's mobile.

26. **Universal Remote Controller (IR)**

    This system is designed to act as a hub for multiple household appliances. It "learns" the Infrared (IR) signals from original remotes (TV, AC, DVD) and stores them. The user can then control all these devices from a single interface, such as a mobile app or a custom-built keypad, eliminating the need for multiple controllers.

27. **Morse Code Transmitter/Translator**

    This project serves as a communication bridge. It features a sender side who inputs morse code, then it is sent to the receiver where the message is decoded and displayed.

28. **Energy Meter Monitoring**

    This system tracks the real-time electricity consumption of a household or specific appliance. It monitors current and voltage to calculate power usage. This data is sent to a mobile or web application, allowing the user to view their consumption history, receive alerts for high usage, and see an estimated cost of their electricity bill in real-time.

29. **Toll Booth Manager**

    This system automates the process of highway toll collection using RFID technology. Each vehicle is equipped with a unique RFID tag. As the vehicle approaches the booth, the system reads the tag, verifies the user's account balance, and automatically deducts the fee. If the balance is sufficient, the gate opens; otherwise, the gate remains closed and a warning signal is triggered.

30. **Smart Library System**

    This system simplifies common library operations such as borrowing and returning. The system should be aware of the current inventory of books and their status (whether they are borrowed by any customer). It should provide real-time book tracking.